

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 09



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Recap of Previous Lecture



Topic

Polyhedral graph

Topic

Euler's equation for disconnected planar graph

Topic

Vertex coloring

Topic

Chromatic number

Topics to be Covered



Topic

Chromatic number



Topic

Welsh Powell's algorithm for vertex coloring



Topic

Matching



Topic

Matching number

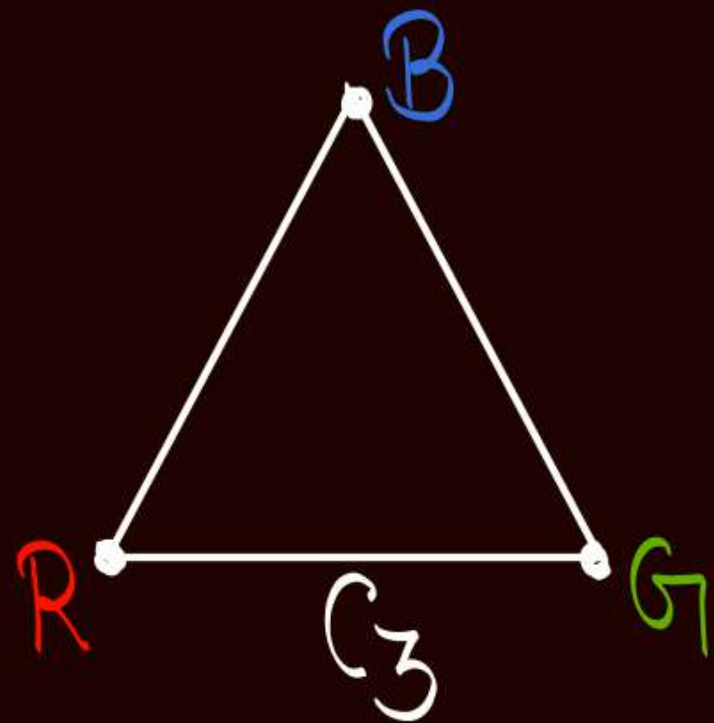


Topic

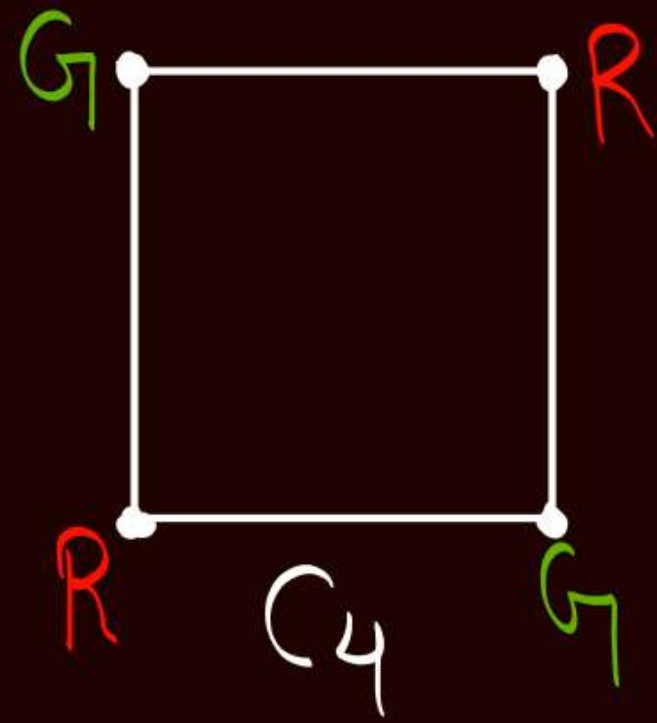
Perfect matching



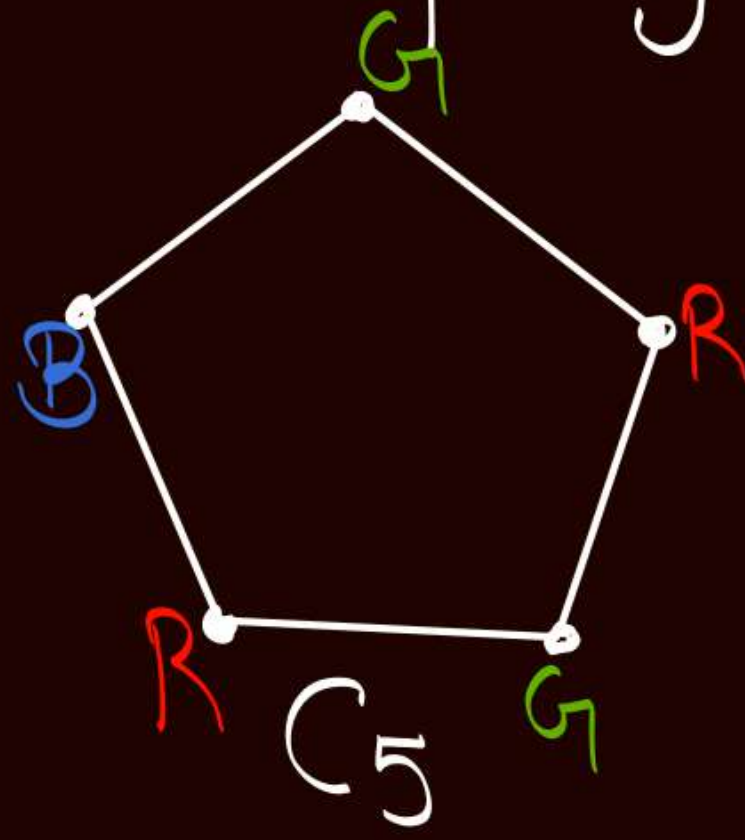
HW Q:- Find the Chromatic number of Cycle graph ' C_n '



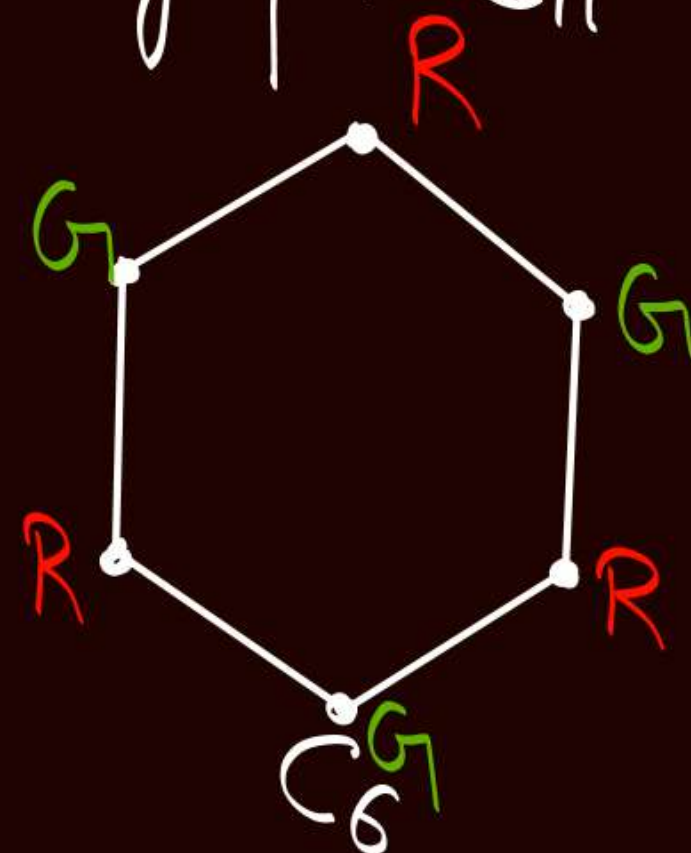
$$\chi(C_3) = 3$$



$$\chi(C_4) = 2$$



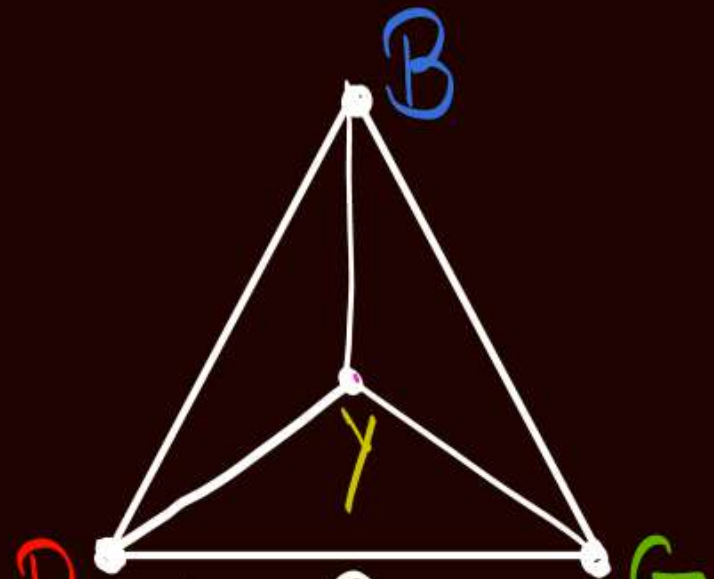
$$\chi(C_5) = 3$$



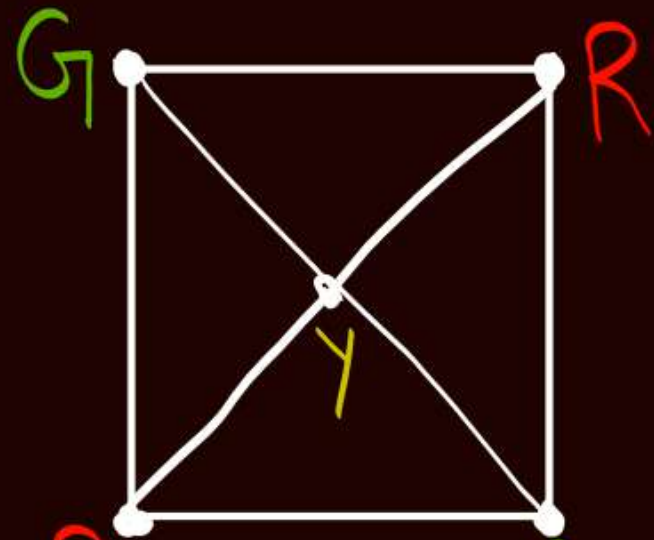
$$\chi(C_6) = 2$$

$$\chi(C_n) = \begin{cases} 2 & \text{if } n = \text{Even} \\ 3 & \text{if } n = \text{odd} \end{cases}$$

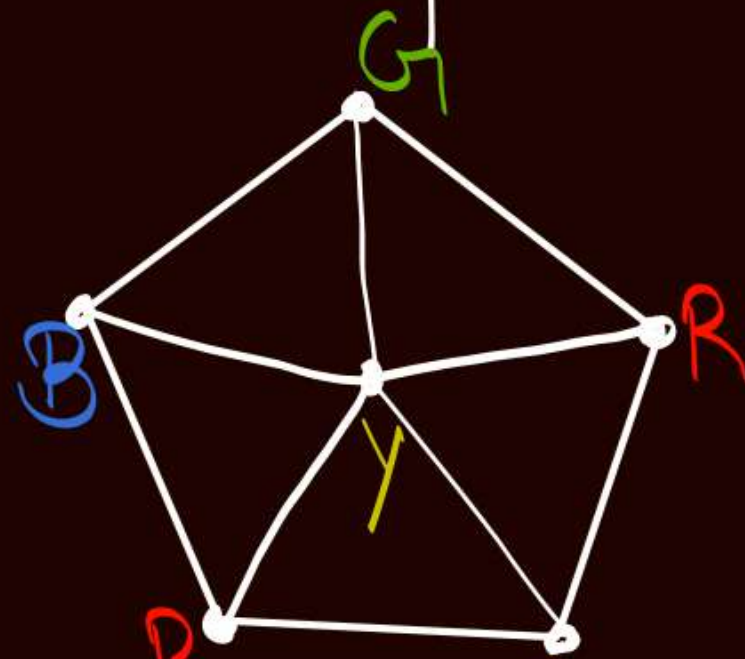
HW Q:- Find the Chromatic number of Wheel graph W_n



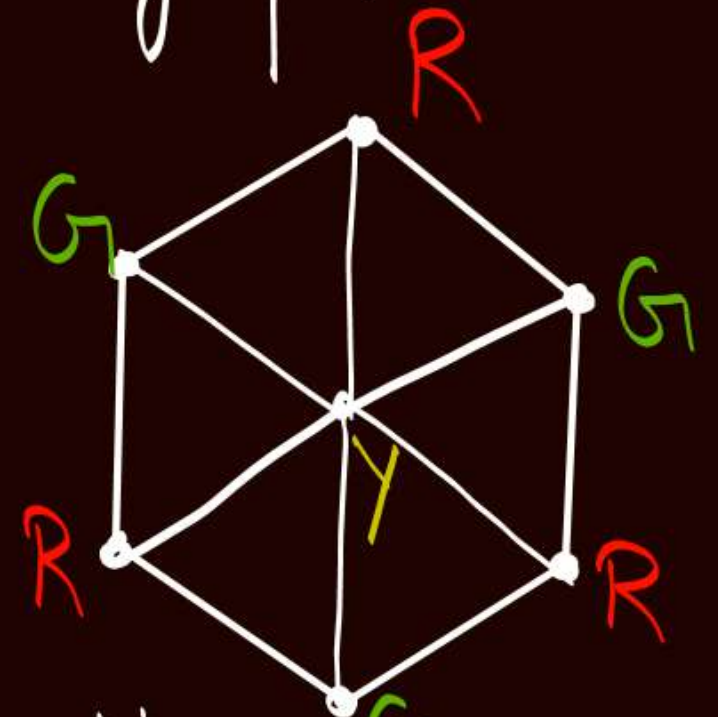
$$W_4 = C_3 + 1 \text{ hub}$$
$$\chi(W_4) = 3 + 1 = 4$$



$$W_5 = C_4 + 1 \text{ hub}$$
$$\chi(W_5) = 2 + 1 = 3$$



$$W_6 = C_5 + 1 \text{ hub}$$
$$\chi(W_6) = 3 + 1 = 4$$



$$W_7 = C_6 + 1 \text{ hub}$$
$$\chi(W_7) = 2 + 1 = 3$$

$$\chi(W_n) = \begin{cases} 4 & \text{if } n = \text{Even} \\ 3 & \text{if } n = \text{odd} \end{cases}$$

H.W.
Q:- Find the Chromatic number of Complete bipartite graph $K_{m,n}$



No two vertices of same subset are adjacent to each other.

∴ All vertices of same subset can be colored using same color

Assign one color to 1st (and) subset

Assign one color to 2nd subset

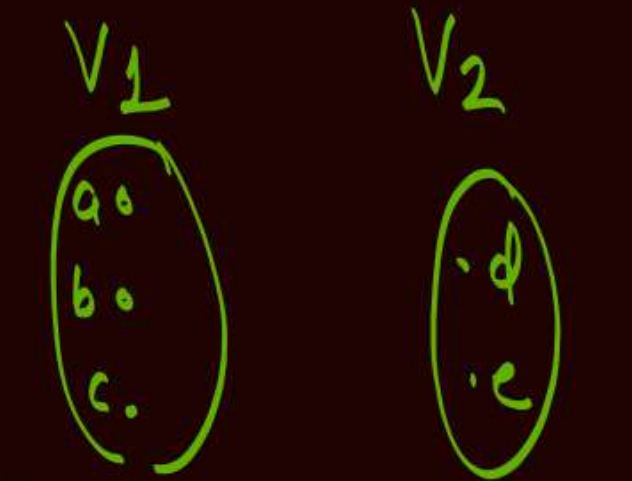
∴ '2' colors are required

$$\chi(K_{m,n}) = 2$$

→ Complete bipartite graph can never be a "null graph" because no subset of partition can be empty.

→ But a bipartite graph may be a NULL graph

eg.



it is a bipartite graph
and it is a NULL graph

★ For any "non-NULL" bipartite graph

$$\chi(\text{Non-NULL bipartite graph}) = 2$$

and for a bipartite graph which is a NULL graph

$$\chi(\text{graph}) = 1$$

∴ Bipartite graphs are also known as
"bi-colorable" graphs {i.e., 2-Colorable}

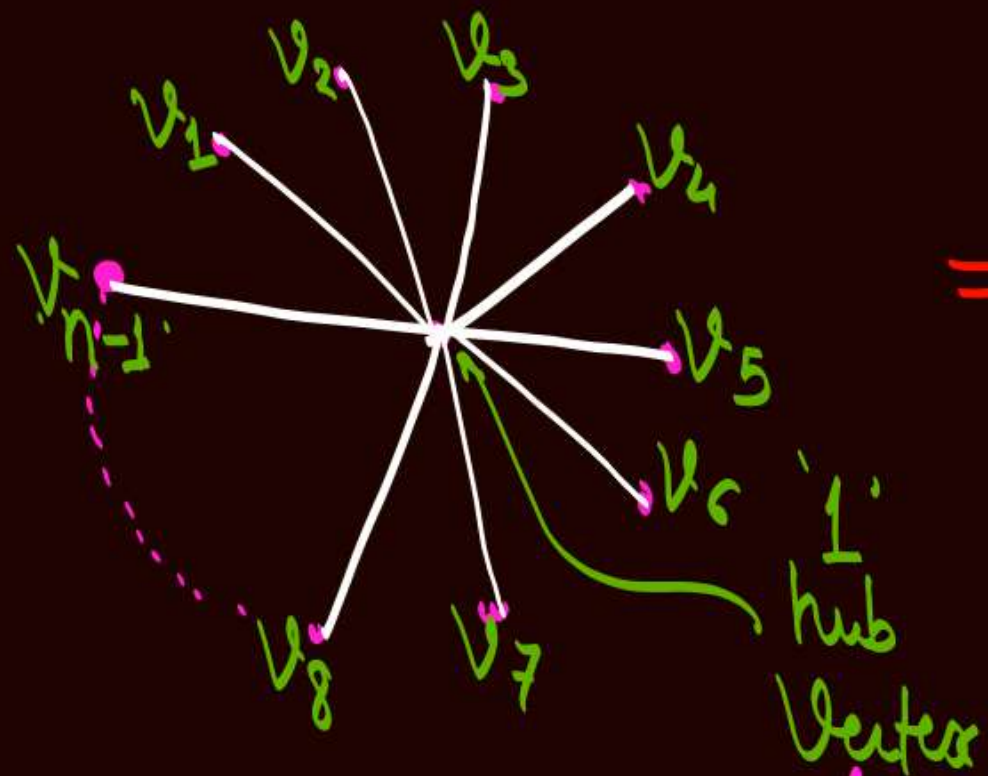
Note:-

If Chromatic number of a graph 'G' is 2,
then graph G is a bipartite graph
i.e., if $\chi(G) = 2$, then 'G' is bipartite

Note:

- ① In a graph G , if there exists at least one cycle of odd length, then $\chi(G) \geq 3$
for cycle itself '3' colors are required
- ② In a non-NULL graph G , if Every cycle is of Even length then $\chi(G) = 2$
{'0' included}
- ③ In a graph G , if all cycles are of even length then graph G is a bipartite graph {provided $|V| \geq 2$ }
- ④ If $\chi(G) = 2$, then ' G ' is a bipartite graph and hence Every cycle in graph G is of Even length
- ⑤ For a non-NULL tree (ie, for any tree with two or more vertices) the Chromatic Number = 2

H.W Q:- Find the Chromatic number of a star graph with 'n' vertices



1 hub vertex
(n-1) vertices
Connected with
hub vertex

Star graph with 'n' vertices

This graph
can be
visualized as

$K_{1, n-1}$

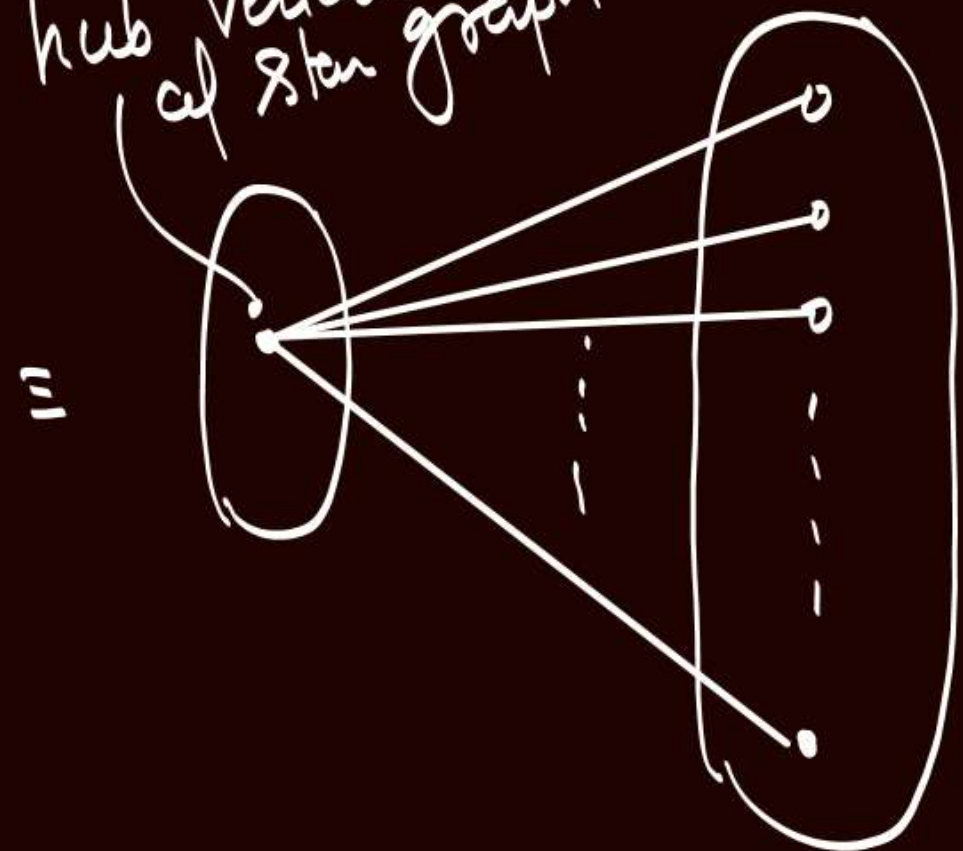
(or)

$K_{n-1, 1}$

$\therefore$

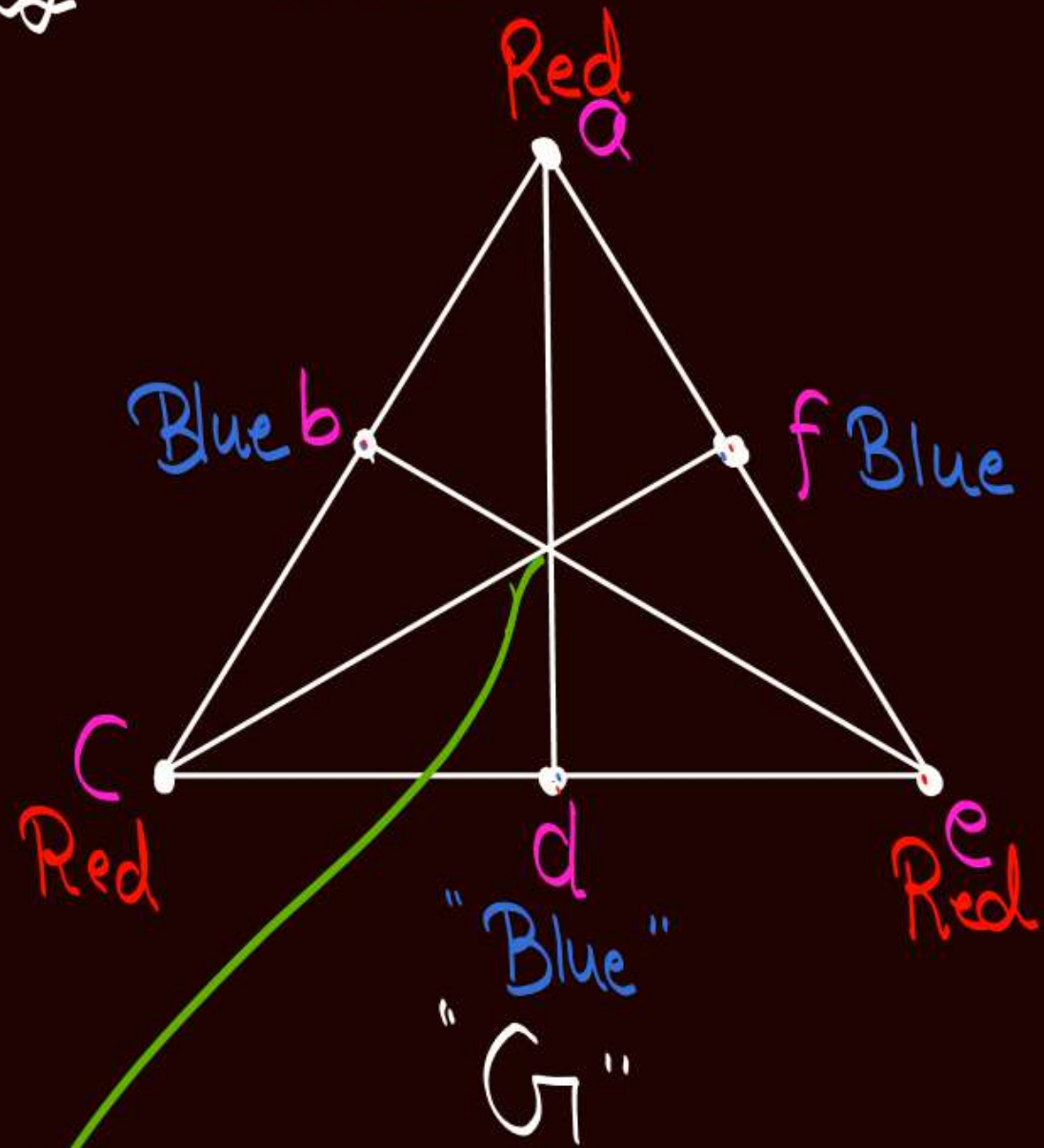
$$\chi(\text{Star graph with } n \text{ vertices}) = 2$$

hub vertex
of star graph



all other
(n-1) vertices
of star graph

eg:- Find the Chromatic number of following graph, "G"



it is not
a vertex

→ We can vertex color the graph G
using '2' colors, $\therefore$ G is "2-Colorable"
i.e., $\chi(G) \leq 2$ ——— eqⁿ ①

→ 'G' is not a NULL graph
 $\therefore \chi(G) \geq 2$ ——— eqⁿ ②

By eqⁿ ① & eqⁿ ②
 $\chi(G) = 2$

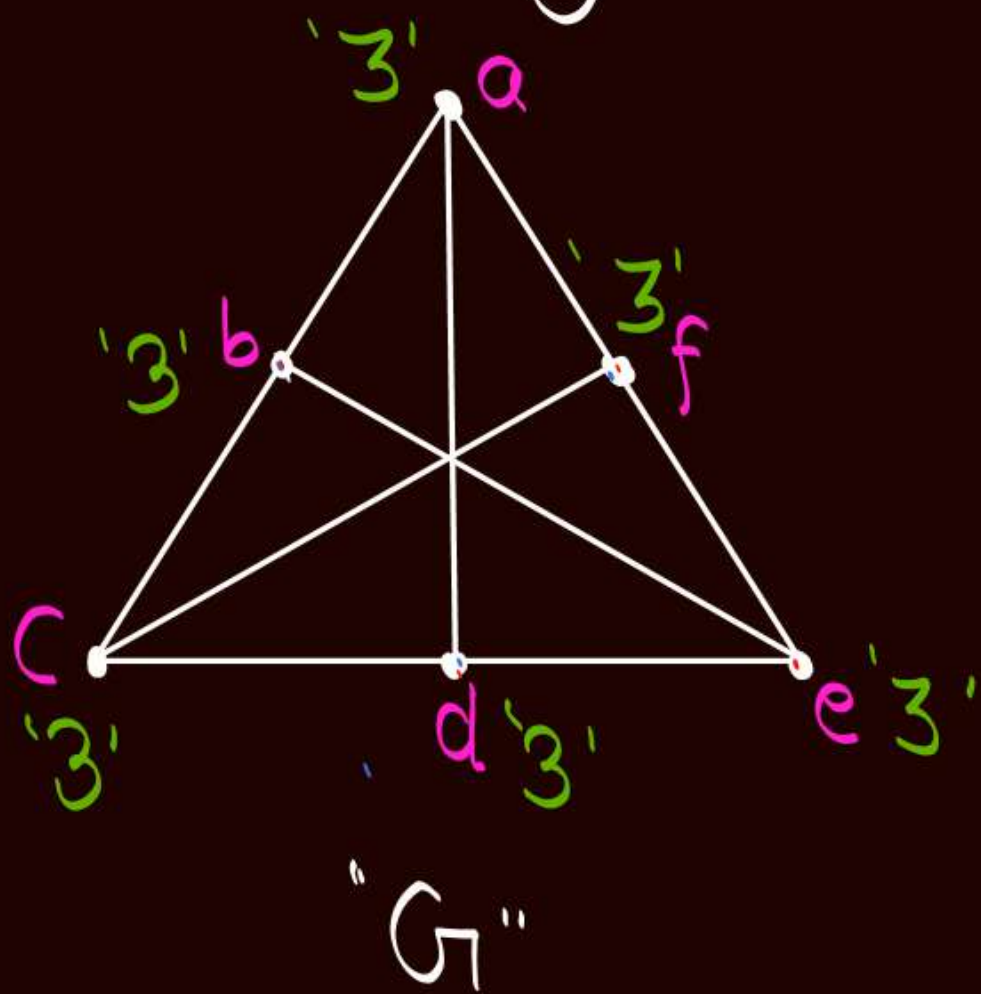


Topic : Welsh Powell's Algorithm

It is a "greedy algorithm", so it need not produce optimal result. 

- It is used to check the upper bound on chromatic number of a graph G.
- 1. Arrange the vertices of the graph G in non increasing order of their degrees.
- 2. If two or more vertices are having the same degree then use alphanumeric order.
- 3. Assign the colors to the vertices in non increasing order of their degrees such that no two adjacent vertices have the same color.
- 4. If Welsh Powell's algorithm uses "m" colors for a graph G, then $\chi(G) \leq m$. {i.e, Graph 'G' is m-Colorable}
'm' is the upper bound on Chromatic number of graph G

eg:- Find the Chromatic number of following graph, "G"
Using "Welsh Powell's algo"



vertices are of same degree
∴ use alphanumeric order

Vertices in
non-increasing
order of their
degrees

Color Used

a

b

c

d

e

f

Red

—

Red

—

Red

—

Green

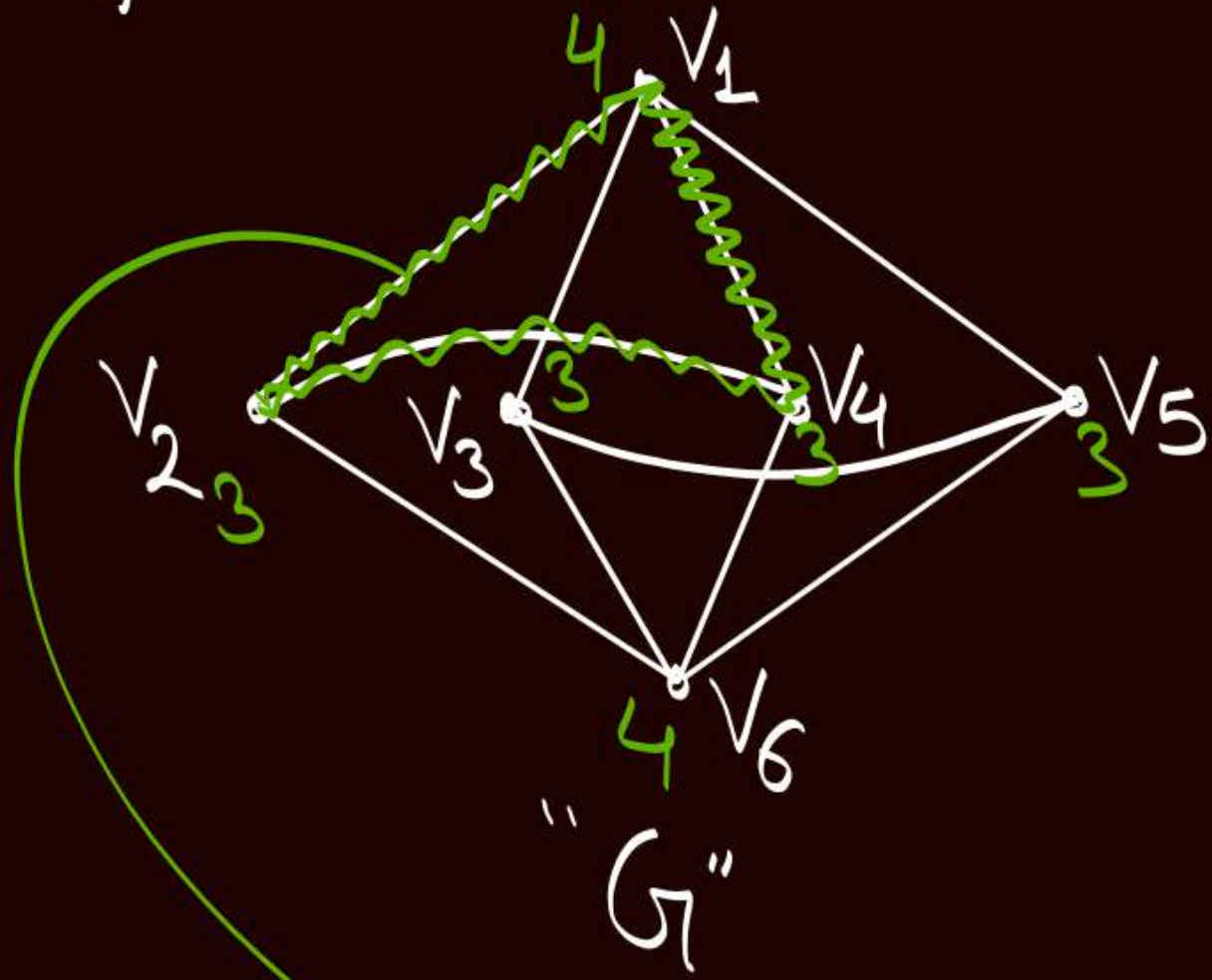
Green

Green

→ Welsh Powell's algorithm uses '2' colors ∴ $\chi(G) \leq 2$ — ①
We know 'G' is not a NULL graph ∴ $\chi(G) \geq 2$ — eq ②

By ① & ② $\chi(G) = 2$

Q:- Find the Chromatic number of the following graph 'G'



Vertices in non-increasing order of deg.	alphanumeric		alphanumeric			
	V1	V6	V2	V3	V4	V5
Colors	Red	Red	— Blue	— Blue	— Green	— Green

Welsh Powell's algo uses '3' colors $\therefore \chi(G) \leq 3$ — eqn ①

There exist some cycles of odd length
 $\therefore \chi(G) \geq 3$ — eqn ②

By ① & ② $\chi(G) = 3$



Topic : Welsh Powell's Algorithm

→ Welsh Powell's algo need not produce optimal result in all cases

eg: Find the Chromatic number of following graph G

Welsh Powell's algo

all vertices are of same degree
∴ Use alphabetic order



G

Vertices	a	b	c	d	e	f
Color	Red	Red	—	—	—	—
			Green	Green		
					Blue	Blue

'3' colors are used by Welsh Powell's algo ∴ $\chi(G) \leq 3$

But it is a bipartite graph
∴ $\chi(G) = 2$

Edge Coloring: - An assignment of colors to the edges of graph G , such that no two adjacent edges are colored using the same color is called "Edge Coloring" of graph G .

- The minimum number of colors required for Edge Coloring of graph ' G ' is called "Index Number" of graph G .



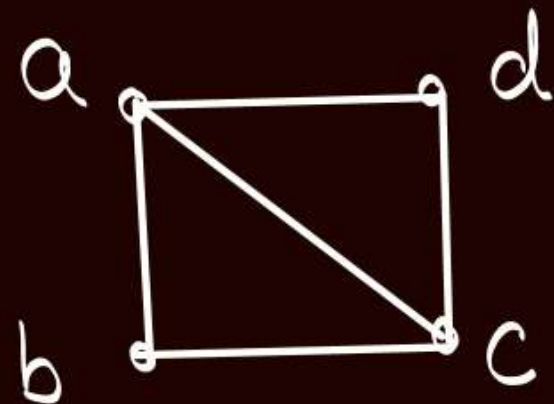
Topic : Matching

A subgraph M of graph G is called a matching of graph G , if all the vertices of graph G are incident with at most one edge in subgraph M ,

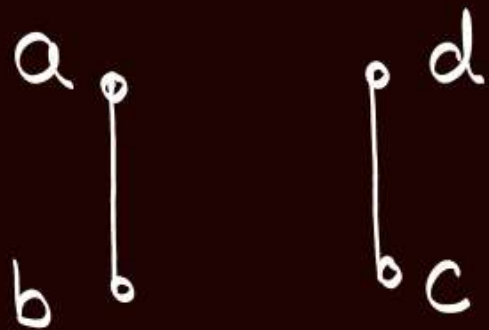
{ i.e. In a matching of graph G ,
all the vertices of graph G must be present }

i.e. In a matching of graph of G $\deg(v) \leq 1, \forall v \in G$

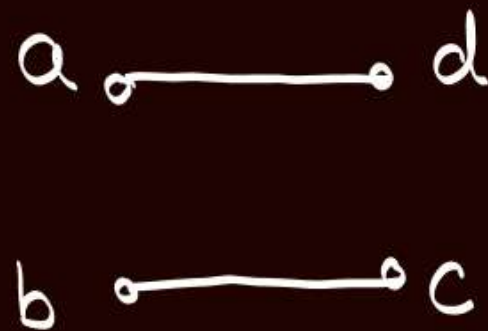
eg:



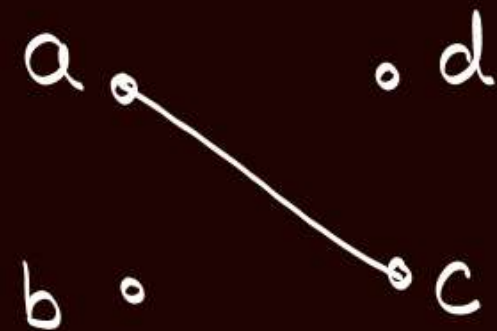
G



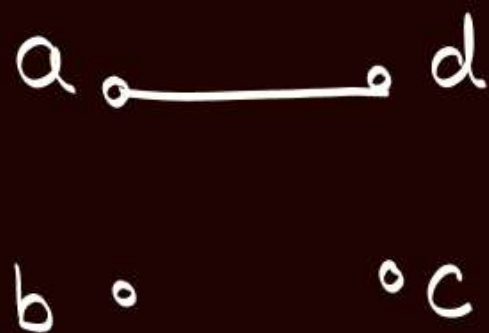
"M₁"



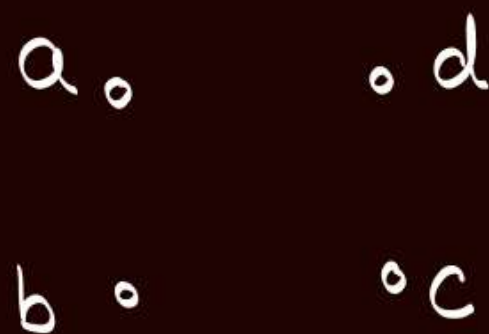
"M₂"



"M₃"



"M₄"



"M₅"

Note: In a matching adjacent edges are not allowed & otherwise $\deg > 1$

M₁, M₂, M₃,
M₄, M₅, ... etc.
all are Matching
of graph G

There are other
Matching as well
for graph G.

Note:

- ① In a Matching of graph G .
if $\deg(v) = 1$, then vertex ' v ' is called a matching vertex
- ② In a Matching of graph G .
if $\deg(v) = 0$, then vertex ' v ' is called a non-matching vertex



2 mins Summary



Topic

Chromatic number

Topic

Welsh Powell's algorithm for vertex coloring

Topic

Matching

Topic

Matching number

Topic

Perfect matching

THANK - YOU